

AD 502 – Machine Learning

Pre-Reading Material – Lecture 4 Batch Learning Vs Online Learning

Estimated Reading Time: 20–25 minutes

Reference Reading: CS4787: *Principles of Large-Scale Machine Learning Systems*, Cornell University. Lecture 26, 2019.

Why This Topic Matters

Machine learning models can learn from data in different ways. In **batch learning**, a model is trained using a fixed dataset and is generally deployed after training without automatically changing its parameters. In **online learning**, the model learns continuously as new examples arrive.

This distinction is important when data is continuously generated or when the underlying patterns may change over time. Applications such as spam detection may need to make predictions while simultaneously adapting to new patterns.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Differentiate between batch learning and online learning.
- Describe the basic workflow of batch learning.
- Explain the prediction–feedback–update cycle of online learning.
- Identify applications where online learning may be useful.
- Explain the basic idea of regret in online learning.



Activate Your Prior Knowledge

1. A changing spam filter

Suppose a spam filter was trained six months ago. Spammers are now using new words and new message patterns.

Think: Should the model continue using its old parameters, or should it learn from new emails?

2. Continuous data

A bank receives thousands of transactions every minute.

Think: Is it practical to completely retrain the model after every new transaction?

3. Learning from feedback

Imagine a system follows this cycle:

Prediction → Actual Result → Update → Next Prediction

Think: How is this different from conventional model training?

Prerequisite Concepts

You should be familiar with:

- Training and testing data
- Features and labels
- Model parameters
- Prediction and prediction error
- Loss function
- Basic idea of gradient descent

Core Concepts

1. Batch Learning

In **batch learning**, the model is trained using a relatively fixed dataset.

A typical workflow is:

Dataset → Preprocessing → Training → Validation/Testing → Deployment

The model learns its parameters from the training data. After deployment, the model generally **does not change automatically**. If new data becomes available, the model may need to be retrained.

Example

A college develops a model to predict student performance using data from previous years.

The model is trained on the historical dataset and then deployed.

If the model remains unchanged after deployment, this is a **batch learning** approach.

2. Online Learning

In **online learning**, learning and prediction are interleaved.

For each new example:

New Example → Prediction → Loss/Feedback → Model Update

The process then repeats for the next example.

For example, a spam detection system may:

1. Receive a new email.
2. Predict whether it is spam.
3. Receive the actual result.
4. Calculate the prediction loss.



5. Update its model.
6. Use the updated model for the next email.

Thus, the model can adapt as new data arrives.

3. Batch Learning vs. Online Learning

Aspect	Batch Learning	Online Learning
Data	Fixed/collected dataset	New data arrives continuously
Training	Before deployment	During operation
Model update	Usually through retraining	Can update as new examples arrive
Adaptation	Relatively slower	Can adapt over time
Suitable for	Stable datasets	Streaming/changing data
Example	Periodic house-price prediction model	Continuously adapting spam filter

Remember

Batch Learning: Learn first → Deploy later

Online Learning: Predict → Learn → Update → Predict again

4. Online Gradient Descent

One algorithm used for online learning is **Online Gradient Descent**.

At each step, the model updates its parameters using the gradient of the current loss:

$$w_{t+1} = w_t - \alpha \nabla w_t \ell(h(w_t(x_t)), y_t)$$



The important idea is that the current example is used to improve the model before the next example is processed.

5. Regret

Online learning evaluates performance using a concept called **regret**.

In simple terms:

Regret represents the extra loss incurred by the online learner compared with a suitable fixed parameter choice.

As more examples are processed, cumulative regret normally increases. Therefore, the goal is not necessarily zero regret, but **sublinear regret**, where regret grows more slowly than the number of examples.

Guided Reading Questions

1. What is the main difference between batch and online learning?
2. Why does a batch-learning model generally need retraining to learn from new data?
3. What are the steps in the online learning cycle?
4. Why is spam detection a suitable application for online learning?
5. What is meant by regret in online learning?

Reflection Activity

Write **100–150 words** on one real-world application where data continuously arrives.

Consider:

- What new data is generated?
- Would batch or online learning be more appropriate?
- Why?

Examples: **banking, healthcare, e-commerce, cybersecurity, social media.**

Self-Assessment

Answer the following before class:

1. In _____ learning, a model is generally trained using a fixed dataset before deployment.
2. In online learning, prediction and _____ are interleaved.
3. A model that updates itself as new examples arrive is using _____ learning.
4. Online learning is particularly useful for _____ data streams.
5. The extra loss incurred by an online learner compared with a fixed parameter choice is called _____.
6. True _____ or _____ False:
A traditional batch-learning model automatically changes whenever new data arrives.
7. Give one real-world application of online learning.

Key Takeaway

Batch learning is appropriate when we can train a model using an available dataset and periodically retrain it when necessary.

Online learning is useful when data arrives continuously and the model needs to adapt while making predictions.

The central question to remember is:

Does the model learn once from a dataset, or does it keep learning as new data arrives?

